Adventure Safety App

MVP Product & Launch Plan

Kerala-first | Android-first | Adventure Discovery + Digital Rope

Working document - October 2026

Core product idea: a free adventure-discovery app that becomes a real-time "Digital Rope" when a group starts an outdoor activity.

Executive Summary

The app is designed as a broad outdoor-adventure companion rather than only a tracking tool. Users can discover activities and places, understand risks, fees, permits, gear needs and preparation steps for free. When users actually go as a group, they can create a team and activate Digital Rope - a phone-to-phone group-safety layer that detects weakening connectivity, falling-behind patterns and separation before the group reaches the destination without a member.

1. Product Positioning

- Not just a map app.
- Not just a family/location tracker.
- Not just a trekking directory.
- Core differentiation: Adventure Discovery + Group Separation Prevention.
- Initial geography: Kerala.
- Initial platform: Android.

Working positioning line: Discover the adventure. Plan it. Go together. Stay together.

2. First-Time User Flow & Authentication

The app should not force registration immediately on launch. The first screen should show the product value before asking the user to create an account.

1. Install and open the app.
2. See the main home preview: adventure categories, tracking/safety preview, location discovery and featured Kerala content.
3. User taps a meaningful action such as choosing Kerala / starting exploration.
4. If not logged in, show Login / Sign Up.
5. After successful authentication, continue directly to the selected destination/activity flow - do not make the user repeat previous choices.
6. Returning logged-in users go directly to Home / last-used region.

Suggested sign-in methods: Google, Apple where applicable, phone number, and email.

3. Free Adventure Discovery Layer

The discovery and planning layer should be free to maximize adoption and make the app useful even for users who never purchase group tracking.

- Trekking
- Hiking
- Forest / Jungle Walk
- Mountaineering
- Rock Climbing
- Camping
- Cycling
- Mountain Biking
- Trail Running
- Group Jogging
- Kayaking
- Rafting
- Waterfall / Nature Trails

Adventure Detail Page

- Overview and difficulty
- Distance, duration and elevation
- Terrain type and best season
- Entry fee / permit / guide requirement
- How to reach, parking and access notes
- Network availability and facilities
- Water availability and emergency contacts
- Common risks, dangerous sections and wildlife considerations
- Recommended shoes, clothing, food, hydration and first aid
- Beginner / family / group suitability
- Preparation checklist and important precautions

4. Plan Adventure Flow

From an adventure detail page, the primary action is:

PLAN ADVENTURE

Then the user chooses:

Solo	With a Group
Continue with normal planning tools and activity information.	Create / join a team, then optionally activate Digital Rope group safety.

5. Group Creation & Team Joining

- Organizer creates a group session.

- Set activity, place, date, expected start/end time and group size.
- Generate a Team Code / QR Code.
- Members join the same session.
- Dashboard confirms member count, e.g. 8/8 members joined.

6. Digital Rope - Core Safety Feature

Digital Rope is the primary technical differentiator. The phone remains the main group-safety node. Nearby phones maintain a chain of connectivity so the leader does not need a direct Bluetooth connection to the last person.

Phone A <-> Phone B <-> Phone C <-> Phone D <-> Phone E

Safety States

State	Meaning
Connected	Normal heartbeat / stable connection.
Weak	Signal quality or heartbeat trend is degrading.
Falling Behind	Sustained separation trend; alert the member and leader.
Separated	Member is no longer part of the Digital Rope chain.
Critical	No reconnection within the escalation window; stop-group / recovery actions become prominent.

Important Engineering Principle

Do not promise an exact Bluetooth distance such as '43.7 m away'. Terrain, ridges, trees, the user's body, pockets and phone hardware can change radio behavior. Use connection trend + missed heartbeats + time + signal degradation to estimate separation risk.

7. Example: Shoelace / Falling-Behind Scenario

7. One member stops to tie a shoelace while the group continues.
8. A short grace period prevents an unnecessary alarm.
9. If the group continues moving away, the member gets: 'Your group is moving away.'
10. The leader gets: '1 member is falling behind.'
11. If the member is not reconnected, the state escalates to Separated / Critical.
12. The group stops before reaching the destination and discovering the person is missing.

8. Last-Known Position & Recovery Actions

- Last known GPS position
- Disconnect time
- Last connected member
- Approximate point on route
- Direction / recent movement where available
- Connection timeline

Suggested actions:

- I'm OK
- Wait for me
- Need Help

- Stop Group
- Locate Last Position
- Send Group Alert

9. Watch & Wearable Strategy

For the MVP, the Digital Rope should remain phone-to-phone. Watches are companion output devices, not mesh relays.

- Phone remains the primary Bluetooth / Digital Rope node.
- Apple Watch / Wear OS can mirror phone alerts and provide vibration.
- A dedicated watch app can be added later for quick actions such as I'm OK / Wait for me / Need Help.
- WHOOP can be treated later as a health/performance data source, not as the main safety alert or mesh device.

10. Free Trial, Rewards & Retention

The initial monetization model should follow 'use first, trust it, then pay'. Each user gets three real group-safety sessions free.

- 3 Digital Rope group sessions free.
- No heavy paywall before the user experiences the safety value.
- After 3 completed sessions: unlock a permanent digital milestone / shareable card.
- Do not label it as government-certified unless there is an actual formal government partnership.

Suggested Milestone Ladder

Milestone	Example Title	Reward Direction
3 adventures	Digital Rope Explorer	Digital badge / share card
5 adventures	Trail Keeper	Digital badge
10 adventures	Rope Guardian	Sponsor coupon / partner reward where available
20+ adventures	Advanced milestone	Limited sponsor-funded physical reward / patch / accessory

Physical gear such as trekking poles, backpacks or electrolyte packs should not be guaranteed to every early user. Use sponsor-funded giveaways, selected milestones, referral campaigns or limited challenges so acquisition cost remains controlled.

11. Monetization Rollout

Phase	Model
Phase 1 - MVP	Discovery + planning free; 3 Digital Rope sessions free; focus on adoption and proof.
Phase 2 - Validation	Introduce group-session pass and/or approximately ₹169/month plan after users trust the feature.
Phase 3 - Organizer	Guide / trekking-club plan: larger groups, rosters, reports, history and organizer tools.
Phase 4 - Partnerships	Sponsorships, affiliate gear offers, tourism/adventure partnerships, selected rewards.

12. MVP Scope - Build Now

- Android app

- Home preview + sensible authentication gate
- Kerala as first region
- 4-5 priority activity categories
- Approximately 20-30 well-researched locations
- Adventure details: risk, gear, permits, fees, preparation
- Group create / join via code or QR
- 3 free Digital Rope sessions
- Connected / Weak / Falling Behind / Separated states
- Group alerts
- Last-known location
- Basic adventure summary
- Digital milestone / badge

Out of Scope for MVP

- India-wide adventure database
- iOS launch
- Dedicated Apple Watch / Wear OS apps
- WHOOP integration
- Satellite integration
- Full marketplace / gear shop
- Large paid advertising campaigns
- Enterprise guide dashboard
- Guaranteed physical gifts for all users
- Complex social network / chat / gamification

13. Current Team Structure

Person / Area	Role	MVP Responsibility
Shad	Technical / cybersecurity / product build	Core app, Digital Rope logic, security, technical decisions
Amalu	Cybersecurity / technical support	Security testing, QA support, privacy / threat review
Ashna	Product research	Location/activity research, risks, fees, permits, gear/preparation content
Marketing	Handled lean/in-house initially	Community outreach, social media, launch content and trek-group relationships

Budget implication: because product research, cybersecurity and much of the technical work are already in-house, the MVP does not need a normal agency-sized development budget.

14. Field Testing & Hiker India Closed Beta

A real trekking partner is more valuable than laboratory-only testing. The proposed testing sequence is:

13. Internal team testing on 3-5 smaller, safer local routes.
14. Fix Bluetooth drops, false alerts, background app behavior, battery drain, GPS delay and notification issues.
15. Share a closed beta build with Hiker India for real group treks.
16. Run intentional controlled scenarios: one person waits behind, two people separate, screen locked, phone in pocket, battery saver on/off, open hill and forest terrain.
17. Collect feedback and technical metrics.

18. Target 20-50 real group sessions before public MVP release.

Beta Metrics to Capture

- Time from separation trend to first warning
- Time to leader alert
- False-alert rate
- Reconnect time
- Heartbeat misses
- Phone model / Android version
- Battery impact
- Group size
- Terrain category
- User understanding of alert states

Safety note: during beta, the app must be described as an experimental companion tool, not the primary safety device. Normal guide procedures, manual headcounts and emergency protocols must continue.

15. MVP Cash Budget

Recommended approach: do not spend the full ceiling up front. Release money in stages based on technical proof.

Expense	Planning Range	Reason
Google Play account	₹2,000-₹3,000	One-time store registration plus FX/tax buffer
Domain / landing page / product email	₹1,500-₹4,000	Trust, support, privacy policy and launch page
Backend / auth / database / notifications	₹0-₹6,000	Start lean; reserve for production usage
Maps / location API	₹0-₹5,000	Usage buffer
Test phones / power banks / accessories	₹5,000-₹20,000	Depends on devices already owned
Internal field testing	₹5,000-₹10,000	Travel, entry fees, food/logistics; Hiker India can reduce this
UI / app-store assets	₹2,000-₹7,000	Can be lower if created in-house
Privacy / terms / legal buffer	₹3,000-₹10,000	Important for location and safety claims
Security / QA tooling	₹0-₹5,000	Mostly open-source / in-house
Small pre-launch marketing tests	₹3,000-₹10,000	Community and creative experiments
Contingency	₹5,000-₹10,000	Unexpected testing / device / cloud costs

Budget Scenarios

Scenario	Budget	Use
Lean Validation	₹30,000-₹40,000	Existing devices, free cloud quotas, limited local testing, organic marketing.
Recommended MVP	₹70,000-₹80,000	More field/device coverage, privacy/legal buffer, launch assets and controlled marketing.
Comfortable Ceiling	Up to ₹1,00,000	Extra devices, wider Kerala testing, more contingency; still avoids agency outsourcing.

16. 8-10 Week MVP Timeline

Period	Main Work
Weeks 1-2	UX/auth flow, Kerala structure, content/data model
Weeks 3-4	Group create/join, Digital Rope prototype, heartbeat/state logic
Weeks 5-6	Background behavior, alerts, last-known location,

	security/QA
Weeks 7-8	Internal real-world testing + Hiker India closed beta
Weeks 9-10	Fixes, store assets, policies, closed beta expansion / public MVP decision

17. MVP Success Criteria

- 20-50 real group sessions completed in beta.
- Falling-behind / separation warnings arrive early enough to be useful.
- Low false-alert rate after threshold tuning.
- Reconnect behavior is understandable and reliable.
- Acceptable battery impact for a typical outing.
- Group leaders understand the dashboard without training.
- Users voluntarily return after free sessions.
- Real evidence that Digital Rope prevents 'we noticed only at the finishing point' situations.

18. Post-MVP Roadmap

- Paid session pass and/or monthly subscription
- Guide / organizer plan
- Sponsor-funded rewards and gear offers
- Dedicated watch apps
- WHOOP / health-data integrations
- iOS
- More Indian states
- Tourism / forest / adventure partnerships
- Incident reports and organizer exports
- Satellite / advanced offline features only after core product maturity

19. Immediate Next Actions

19. Freeze MVP scope and stop adding nonessential features.
20. Complete PLAN.md with the product, monetization, field-test and budget sections.
21. Build the Android group-flow prototype first.
22. Prepare 20-30 Kerala adventure records with validated data.
23. Create the first internal field-test checklist.
24. Contact Hiker India only after the internal build is stable enough for a controlled beta.
25. Review results after the first 5-10 group tests before spending further budget.

MVP principle: prove that the app can warn a group before a member is truly lost.